

Abhishek Saini

Senior Software Engineer

✉ abhisaini880@gmail.com

☎ +91-8368659459

🌐 www.abhisaini.in

Senior Software Engineer with 7+ years of experience designing and scaling backend systems in Python. Hands-on experience building API-driven services, integrating LLMs into production workflows, and developing internal platforms for automation and productivity. Strong foundation in distributed systems, cloud-native infrastructure, and cost-aware system design, with experience owning services end-to-end, improving performance and efficiency, and collaborating with cross-functional teams to solve real-world operational problems.

Skills

- **Languages & Frameworks**
 - Python (FastAPI & Flask)
 - RESTful API development
- **Data & Storage**
 - MySQL & PostgreSQL
 - Elasticsearch & Redis
 - Snowflake
- **Infrastructure & Platform**
 - AWS
 - Docker & Kubernetes(EKS)
 - Production monitoring and logging
 - CI/CD awareness
- **AI & GenAI**
 - Integrating LLMs into backend services
 - Designing structured, multi-step LLM-driven workflows with deterministic backend logic
 - Building stateful agent workflows using LangGraph
- **System Design & Engineering**
 - Designing scalable backend systems
 - Multi-service and distributed system fundamentals
 - Clean code and maintainable architecture practices
 - Infrastructure cost optimization

Experience

Senior Software Engineer

McKinsey & Company

June 2025 - Present

- **Built agent-based workflows using LLM function calling** to integrate an AI assistant with backend systems for administrative tasks such as meeting scheduling, room booking, and timesheet management.
- **Designed and integrated proof-of-concept flows into production systems**, connecting AI-driven workflows with existing internal services and external APIs.
- **Developed backend services in Python (FastAPI)** with integrations across PostgreSQL (ORM), Slack SDK, Outlook APIs, and webhook-based event subscriptions.
- **Implemented a decision-visualization layer** over the scheduling and room-booking algorithm, exposing intermediate inputs, scoring, and outputs to improve debuggability and explainability of system behavior.
- Worked on scheduling logic that evaluates multiple constraints, including participant availability, existing calendar events, PTOs, and resource availability, to determine optimal time slots.
- **Built an end-to-end data labeling platform** required for evaluation workflows, using Streamlit for UI and Snowflake for data storage and analysis.
- Collaborated with a globally distributed team on feature design, problem-solving sessions, and iterative delivery of product capabilities.
- **Contributed to system reliability** by handling validation, edge cases, and error scenarios across AI-driven and deterministic backend workflows.

Lead Software Developer

Tyroo

2020 - Present

- **Designed robust system architectures**, executed tasks seamlessly, and conducted thorough code reviews to ensure high-quality, maintainable code.
- **Increased code coverage of unit tests to 40%**, enhancing software reliability.
- **Achieved a 45% reduction in AWS costs** through proactive monitoring and optimization.
- Initiated a documentation process achieving 60% coverage, improving knowledge sharing and onboarding.
- **Solely managed the Product Information Management (PIM) service**, processing 10 million products daily, and enhanced product catalog management service capacity from 600 MB to 10 GB using distributed processing, improving scalability.
- Simplified the image creation service communication flow, enhancing performance by 10x and reducing costs by 30%.
- **Mentored a team of 5 developers**, fostering expertise in scalable architecture and coding best practices, resulting in a 30% reduction in onboarding time and improved team performance.
- **Replaced the existing Kong gateway with a custom FastAPI-based gateway**, integrating authentication, rate limiting, and traffic monitoring. Achieved a 25% improvement in API response times, enhanced security, and gained detailed traffic insights through custom metrics.
- Delivered a dynamic video ads feature, transitioning from static image ads, resulting in a 3x improvement in ad performance and engagement rates, driving higher ROI for clients.
- Enhanced system stability by separating staging and production environments in EKS clusters, improving the reliability of production releases and system stability.

Software Engineer

Triotech Solutions

2018 - 2020

- **Automated various OS level and application level tasks** using shell and Python scripting, improving operational efficiency and reducing manual effort.
- Automated reports generation from raw data using Python scripting, streamlining data processing and report generation workflows.
- Developed scripts to increase system efficiency and **reduce human intervention by up to 70%**, optimizing system operations and reducing operational costs.

Education

● Dronacharya College of Engineering (MDU Rohtak)

2018

- Bachelor of Technology, Computer Science